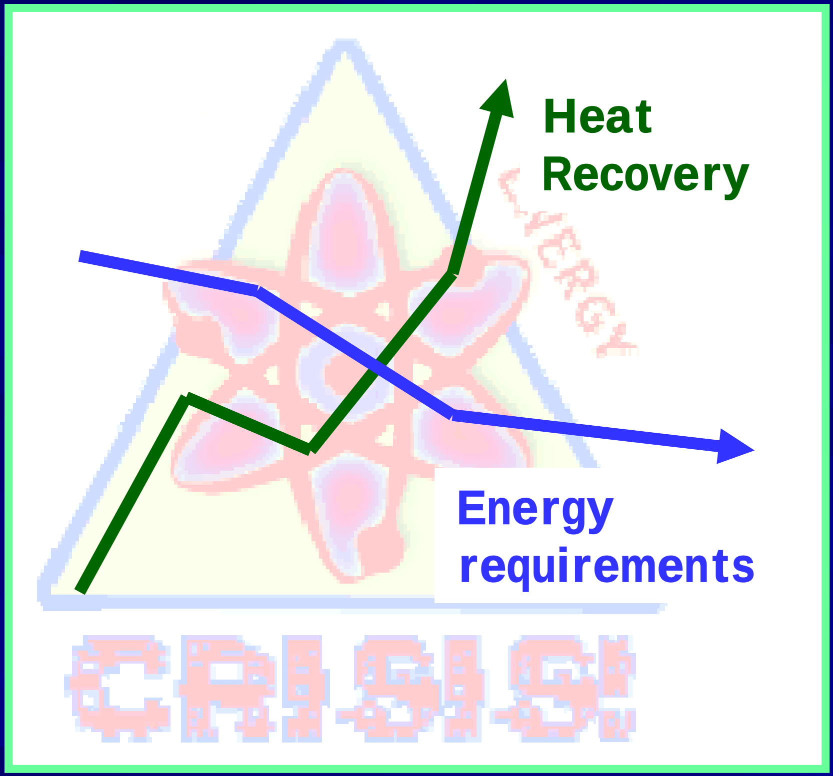
# 1.1 Introduction.

One of the main advantages of Pinch Technology over conventional design methods is the ability to set energy and capital cost targets for an individual process or for an entire production site ahead of design. Therefore, ahead of identifying any specific project, we know the scope for energy savings and investment requirements.

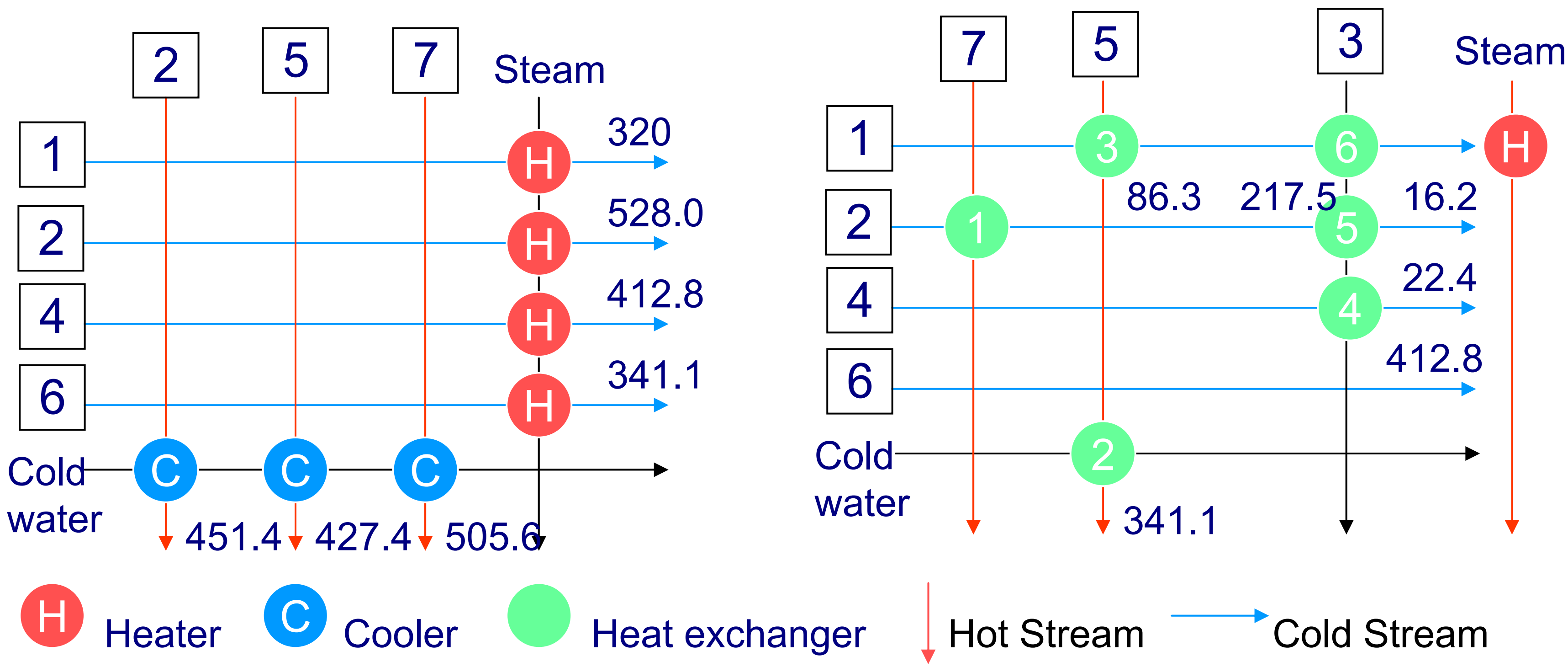
Most industrial processes involve transfer of heat either from one process stream to another process stream (interchanging) or from an utility stream to process stream.

# What is industry challenged about energy consumption and recovery?

In the present energy crisis scenario all over the world, the target of any industrial process designer is to maximizes the process-to-process heat recovery and to minimize the utility (energy) requirements.



To meet the goal of maximize energy recovery or minimum energy requirement (MER) an appropriate heat exchanger network (HEN) is required.




a) Traditional design:
Cost operating 250,838 $/year
Cost capital 4,937 $/year

b) Technology Pinch approach:
Cost operating 24,077.00 $/year
Cost capital 4,180.00 $/year

Fig. 1.1 (a) The non-integrated solution, (b) The optimally integrated solution Reference.

# General Process Improvements

In addition to energy conservation studies, Pinch Technology enables process engineers to achieve the following general process improvements:

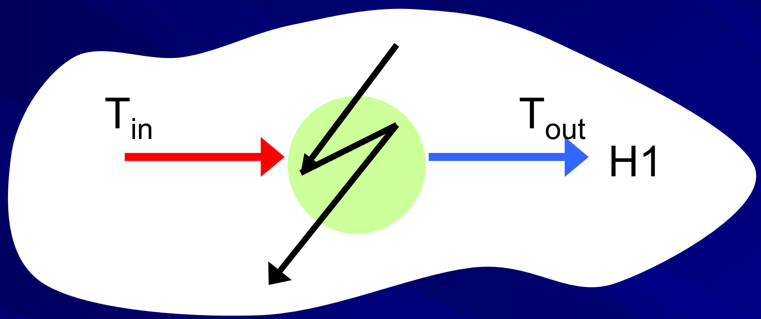
- **Update or Modify Process Flow Diagram**: Pinch quantifies the savings available by changing the process itself. It shows where process change reduce the overall energy target, not just local energy consummation.
- **Conduct Process Simulation Studies**: Pinch replace the old energy studies with information that can be easily updating using simulation. Such simulation studies can help avoid unnecessary capital costs by identifying energy savings with a smaller investment before the projects are implemented.
- **Set Practical targets:** By taking in account practical constrains (difficult fluids, layout, safety, etc.), theoretical targets are modified so that they can be realistically achieved. Comparing practical with theoretical targets quantifies opportunities “lost” by constraints - a vital insight for long term development.
- **De-bottlenecking:** Pinch analysis when specifically applied to debottlenecking studies, can lead to the following benefits compared to a conventional revamp:
  - Reduction in capital costs.
  - Decrease in specific energy demand giving a more competitive production facilities.

# 1.2 Basic Concepts

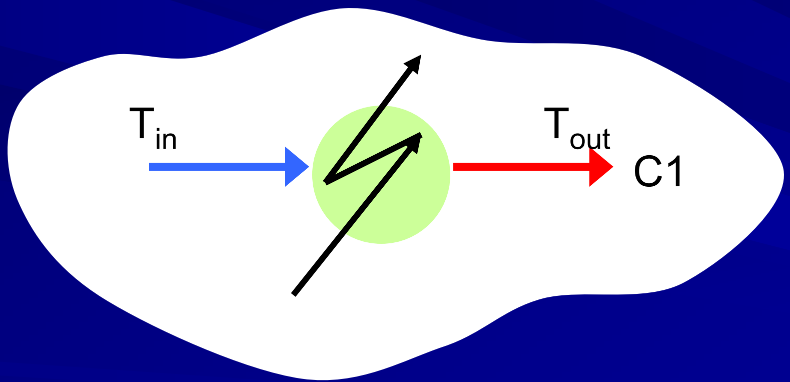
1. Identification of the hot, cold and utility streams in the process.
2. Thermal data extraction for process and utility stream.
3. Selection of Initial $\Delta T_{MIN}$ value.
4. Construction of Composite Curves and Grand Composite Curve.

## 1 Identification of the hot, cold and utility streams in the process.

- **Hot streams:** are those that must be cooled or are available to be cooled ($T_{out} < T_{in}$).




- **Cold streams:** are those that must be heated ($T_{out} > T_{in}$).




- **Utility streams:** are used to heat or cool process stream, when heat exchange between process stream is not practical or economic. A number of different **hot utilities** (steam, hot water, flue gas, etc) and **cold utilities** (cooling water, air, refrigerant, etc.) are used in industry.

## 2 Thermal data extraction for process and utility stream.

For each hot, cold and utility stream identified, the following thermal data is extracted for the process material and heat balance flow sheet:

- Supply temperature $T^S$, the temperature which the stream is available.
- Target temperature $T^T$, the temperature the stream must be taken to.
- Heat capacity flow rate (CP), the product of flow rate and specific heat.
- Enthalpy change H, $H = CP(T^S - T^T)$

| Stream Number | Stream name | Supply Temperature (°C) | Target Temperature (°C) | Heat Capacity Flow Rate (kW/°C) | Enthalpy Change (kW) |
|---|---|---|---|---|---|
| 1 | Feed | 60 | 205 | 20 | 2900 |
| 2 | Reactor out | 270 | 160 | 18 | 1980 |
| 3 | Product | 220 | 70 | 35 | 5250 |
| 4 | Recycle | 160 | 210 | 50 | 2500 |

Table 1.1 Typical Stream Data

## 3 Selection of Initial $\Delta T_{MIN}$ value.

The design of any heat transfer equipment must always adhere to the second law of thermodynamics that prohibits any temperature crossover between the hot and the cold stream I.e. a minimum heat transfer driving force must always be allowed for a feasible heat transfer design.

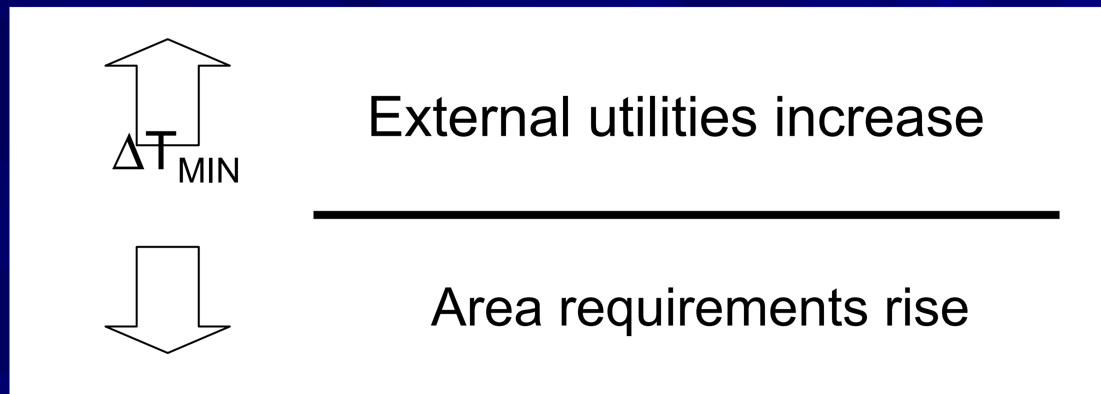
Thus the temperature of the hot and cold stream at the any point in the exchanger must always have a minimum temperature difference ($\Delta T_{MIN}$). This $\Delta T_{MIN}$ value represents the bottleneck in the heat recovery.

In mathematical terms, at any point in the exchanger

$$\text{Hot stream temperature } (T_H) - \text{Cold stream temperature } (T_C) = \Delta T_{MIN} \tag{1.1}$$

The value of $\Delta T_{MIN}$ is determined by the overall heat transfer coefficient (U) and the geometry of the exchanger. In a network design, the type of heat exchanger to be used at the pinch will determine the practical $\Delta T_{MIN}$ for the network.

For a given value of heat transfer load (Q) the selection of $\Delta T_{MIN}$ values has implications for both capital and energy costs.




A few values of based Linnhoff March's application experience are tabulated below for shell and tube heat exchangers.

| No | Industrial Sector | Experience ΔTmin values |
|---|---|---|
| 1 | Oil Refining | 20 – 40 °C |
| 2 | Petrochemical | 10 – 20 °C |
| 3 | Chemical | 10 – 20 °C |
| 4 | Low Temperature Process | 3 – 5 °C |

Table 1.2 Typical ΔTmin values.

# 4 Construction of Composite Curves and Grand Composite Curve.

## Composite Curves

Composite Curves consist of temperature (T) - Enthalpy (H) profiles of heat availability in the process (the Hot Composite Curve) and heat demands in the process (the Cold Composite Curve) together in a graphical representation.

In general any stream with a constant heat capacity (CP) value is represented on a diagram by a straight line running from stream supply temperature to stream target temperature. When there are a number of hot and cold composite curves simply involves the addition of the enthalpy changes of the stream in the respective temperature intervals.

An example of hot composite curves is shown in Fig. 1.2

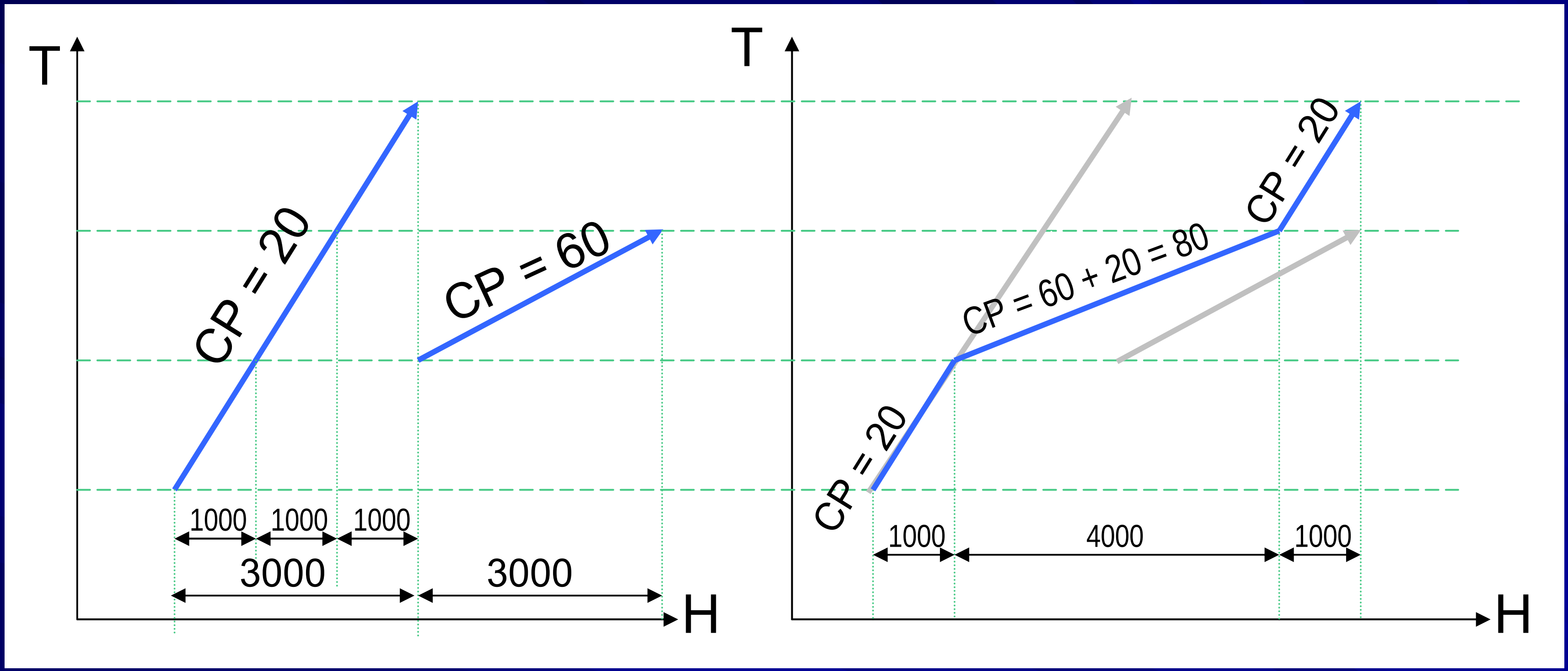



Fig. 1.2 Temperature - Enthalpy relation used to construct Composite Curves.

A complete hot or cold composite curves consists of a series of connected straight lines, each change in slope represents a change in overall hot stream heat capacity flow rate (CP).

## Combined Composite Curves.

Combined Composite Curves are used to predict targets for;

- Minimum energy (both hot and cold utility) required.
- Minimum network area required, and
- Minimum number of exchangers units required.

For heat exchange to occur from the hot stream to the cold stream, the hot stream cooling curve must lie above the cold stream-heating curve.

Because of the "kinked" nature of the composite curves, they approach each other most closely at one point defined as the minimum approach temperature ($\Delta T_{MIN}$).

$\Delta T_{MIN}$ can be measured directly from the T-H profiles as being the minimum vertical difference between the hot and cold curves. *This point of minimum temperature difference represents a bottleneck in heat recovery and is commonly referred to as the "Pinch".*

## $\Delta T$ min and pinch Point.

The $\Delta T_{min}$ values determine how closely the hot and cold composite curves can be "pinched" (or "squeezed) without violating the second law of Thermodynamics (none of the heat exchangers can have a temperature crossover).

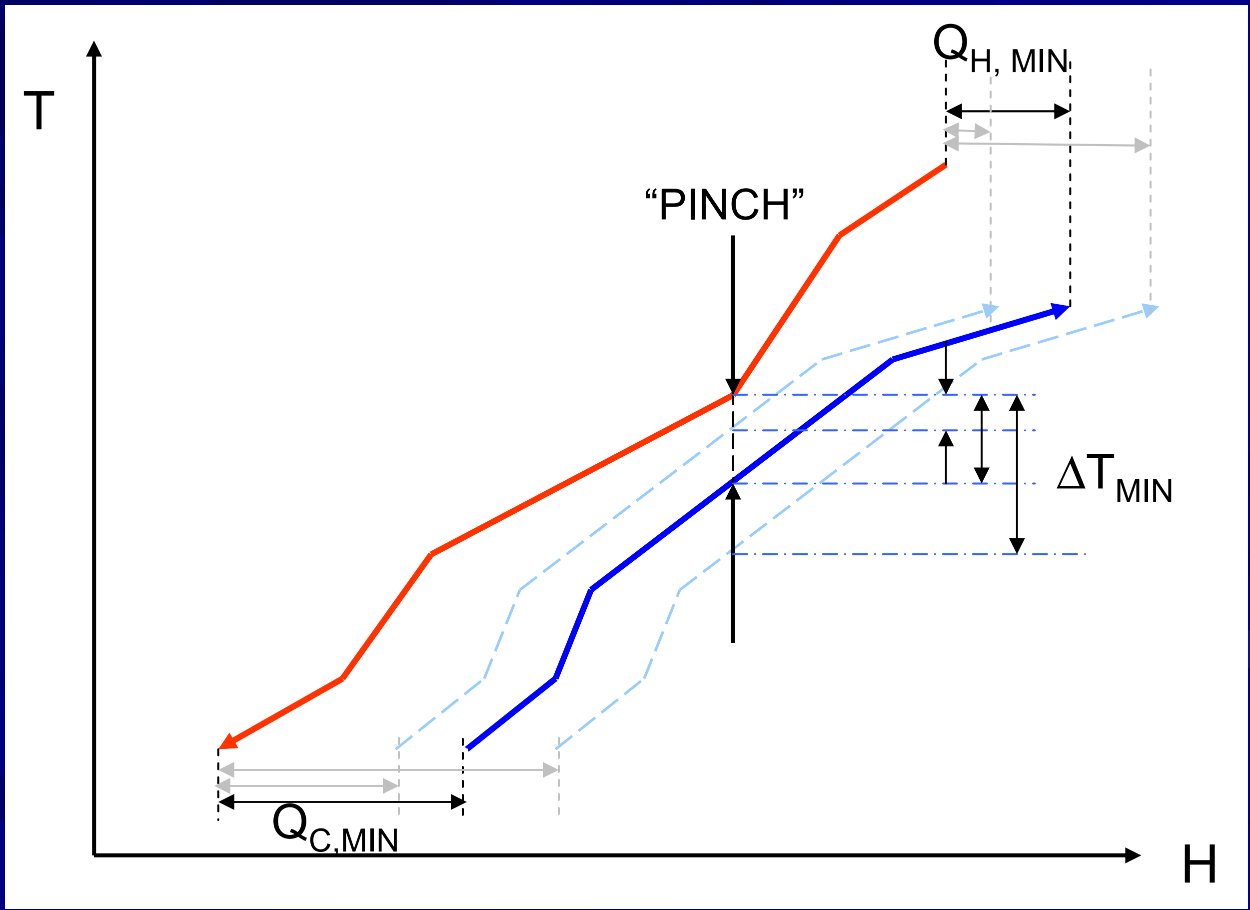



Fig. 1.3 Energy targets and "the pinch" with Composite Curves.

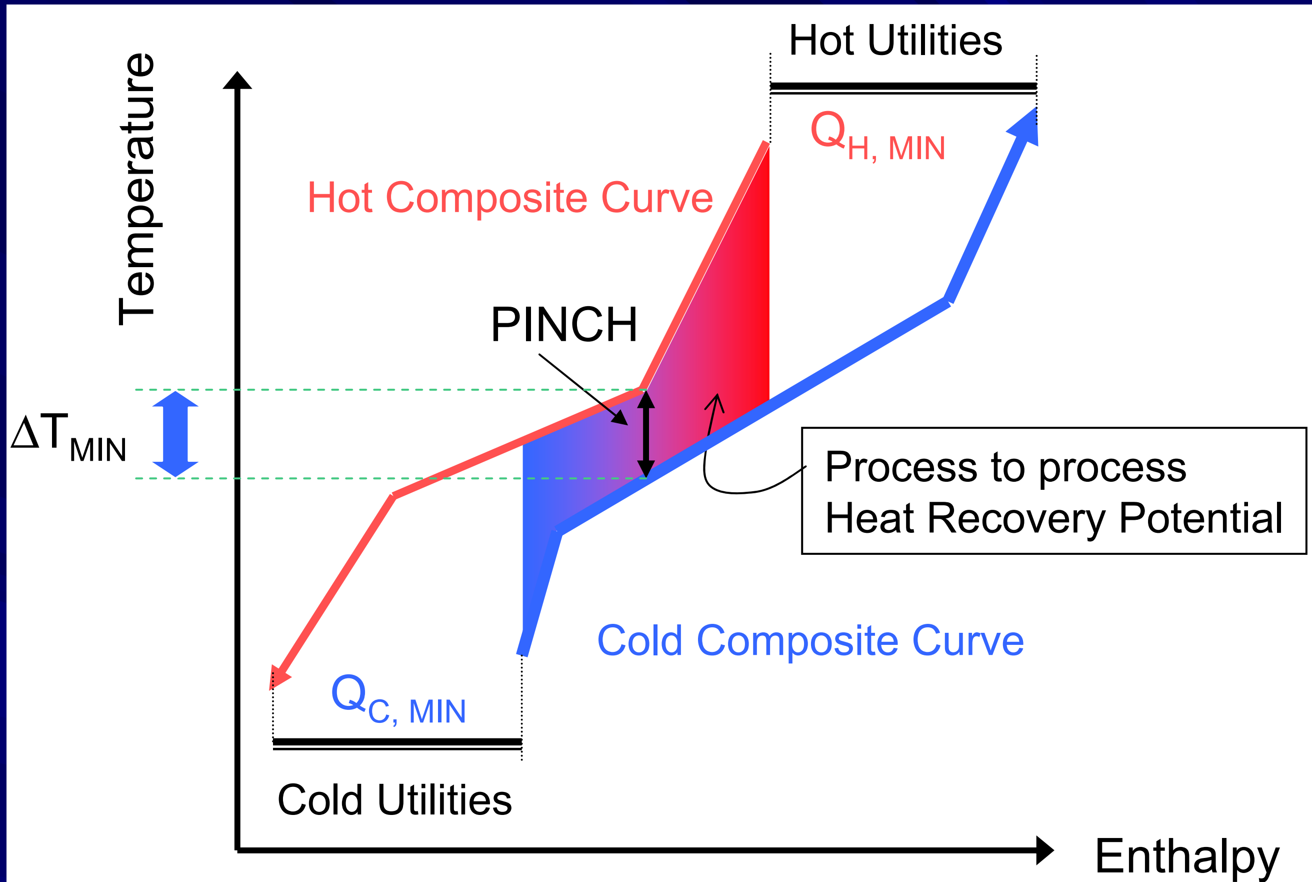



Fig. 1.4 Combined Composite Curves.

At a particular $\Delta T_{MIN}$ value, the overlap shows the maximum possible scope for heat recovery within the process. The hot end and cold end overshoots indicate minimum hot utility requirement ($Q_{H,MIN}$) and minimum cold utility requirement ($Q_{C,MIN}$), of the process for the chosen $\Delta T_{MIN}$.

Thus, the energy requirement for a process is supplied via process to process heat exchange and/or exchange with several utility levels (steam levels, refrigeration levels, hot oil circuit, furnace flue gas, etc.)